

SQL Practice Sources

Curated links for Baraa's 30-Hour SQL Course — organized by topic

All links are free unless noted. Click any link to open it (or copy into your browser). Sources are grouped to match the course structure: Beginner, Intermediate, Window Functions, Advanced, Projects, and Sample Databases.

BEGINNER — SELECT, DDL, DML

w3resource SQL Exercises

<https://www.w3resource.com/sql-exercises/>

500+ exercises across categories (Retrieve data, Operators, Wildcards, Aggregates, Joins, Subqueries), each with solutions.

SQLBolt

<https://sqlbolt.com>

Free interactive lessons + exercises, browser-based, no setup. Great for absolute beginners.

SQL DDL Exercises and Solutions (Scribd)

<https://www.scribd.com/document/895653357/SQL-DDL>

CREATE, ALTER, constraints — with worked solutions.

SQL-Exercises-With-Solutions (GitHub)

<https://github.com/engineeringwitharavind/SQL>

Exercises + solutions organized by topic, MS SQL Server focused.

Educative: Learn DDL Through Practice

<https://www.educative.io/courses/a-practical-guide-to-sql-and-database-fundamentals/learn-ddl-through-practice>

Guided CREATE DATABASE / schema / constraints tasks.

Educative: Learn DML Through Practice

<https://www.educative.io/courses/a-practical-guide-to-sql-and-database-fundamentals/learn-dml-through-practice>

INSERT / UPDATE / DELETE hands-on tasks.

INTERMEDIATE — Filtering, Joins, Functions, Aggregates

w3resource: Joins on HR Database

<https://www.w3resource.com/sql-exercises/joins-hr/index.php>

27 exercises specifically on JOIN types using the HR schema.

100 SQL Exercises (jadedbono, GitHub)

https://github.com/jadedbono/hundred_sql_exercises

100 exercises in ascending difficulty on the HR schema; notes, hints, and solutions included.

LeetCode SQL 50 Study Plan

<https://leetcode.com/studyplan/top-sql-50/>

Official free study plan: Select, Joins, Aggregate Functions, Sorting/Grouping, String Functions.

SQL Exercise Repository (AvinashDarekar, GitHub)

<https://github.com/AvinashDarekar/MY-SQL>

Folder-per-topic exercises: WHERE, CREATE/INSERT, JOINS, ORDER BY, Math Functions.

WINDOW FUNCTIONS

LeetCode SQL 50 (Window/Ranking section)

<https://leetcode.com/studyplan/top-sql-50/>

Dedicated problems using ROW_NUMBER, RANK, DENSE_RANK, PARTITION BY.

01.SQL — WindowFunctions folder (anshuldass, GitHub)

<https://github.com/anshuldass/01.SQL>

Questions/Solutions split into Beginner → Expert (Data Engineering) levels, built on AdventureWorks.

PostgreSQL Exercises (PGExercises)

<https://pgexercises.com>

80+ exercises progressing from SELECT to window functions and recursive queries, one clean dataset.

ADVANCED — Subqueries, CTEs, Views, Procedures, Triggers, Indexes

PostgreSQL Exercises (PGExercises)

<https://pgexercises.com>

Best single source for subqueries, joins, aggregations, and recursive CTEs on one dataset.

01.SQL — CTEs & StoredProcedures folders (GitHub)

<https://github.com/anshuldass/01.SQL>

Questions/Solutions folders for CTEs and Stored Procedures, 2-4 difficulty levels each.

SQL-for-beginners: Oracle PL/SQL Labs (GitHub)

<https://github.com/CodingwithSanta/SQL-for-beginners>

10 labs covering DDL/DML, stored procedures, functions, triggers, exception handling — full solutions.

SQL coding practice (vallianand, GitHub)

<https://github.com/vallianand/SQL-coding-practice>

Single-file practice set including views, stored procedures, and a trigger example (AFTER DELETE log).

Optimization of SQL Query Tuning and Performance (Coursera)

<https://www.coursera.org/projects/optimization-of-sql-query-tuning-and-performance>

Free-to-audit guided project: EXPLAIN/EXPLAIN ANALYZE, indexes, join strategies (~2 hrs).

Database Tuning labs (getintogit1, GitHub)

<https://github.com/getintogit1/database-tuning>

6 labs: query tuning, index theory, index practice, join tuning, concurrency.

PROJECTS — Data Warehouse, EDA, Analytics

Baraa's Free Course Materials (Substack)

<https://datawithbaraa.substack.com/p/access-to-course-materials>

Official links to the SQL Data Warehouse Project and SQL Data Analytics Project — the exact companion projects for this course.

SQL Advanced EDA (kurushika, GitHub)

<https://github.com/kurushika/SQL---Advanced-EDA>

Numbered SQL scripts: database exploration, dimensions, dates, measures, magnitude, ranking — mirrors Baraa's project structure.

Data Cleaning and EDA using SQL (vasukrishna001, GitHub)

https://github.com/vasukrishna001/DataCleaning_and_EDA_usingSQL

Realistic messy-data cleaning + EDA project using a layoffs dataset.

SAMPLE DATABASES — to practice against

Sakila (MySQL official)

<https://dev.mysql.com/doc/sakila/en/>

Free, widely used sample DB; enough complexity for joins, subqueries, window functions.

Northwind (Microsoft, GitHub)

<https://github.com/microsoft/sql-server-samples>

Classic, lighter-weight sample DB — good for beginner to intermediate practice.

AdventureWorks (Microsoft, GitHub)

<https://github.com/Microsoft/sql-server-samples/releases>

Heavier sample DB — best for advanced / window function / CTE practice.

PGExercises Club Dataset

<https://pgexercises.com>

Small, clean, purpose-built dataset that ships with the PGExercises practice site.